Forces

SPH-3UI Grade 11 Physics

April 24, 2026

What is Force?

A force is a **push** or a **pull** that can cause an object to change its motion or shape.

In physics, force is represented by a **vector** because it has both magnitude and direction. For example, pushing a book sideways will have a different effect than pulling it upward.

Forces are the fundamental way that matter interacts with matter.

Everyday Forces

Example: Imagine two children playing outside with a wagon. One child is pulling forward on a rope tied to the front, while the other pushes from behind.

Question: Which forces act on the wagon?

Applied Force ($\vec{F}_{app}$)

An **applied force** results when one object is in contact with another and pushes or pulls on it.

- Example: The child behind the wagon exerts an applied force forward on the wagon by pushing on the back.
- Symbol: $\vec{F}_{app}$
- Direction: Points away from the surface of contact.

Tension ($\vec{F}_T$)

Tension is a pulling force exerted on an object by a rope or string.

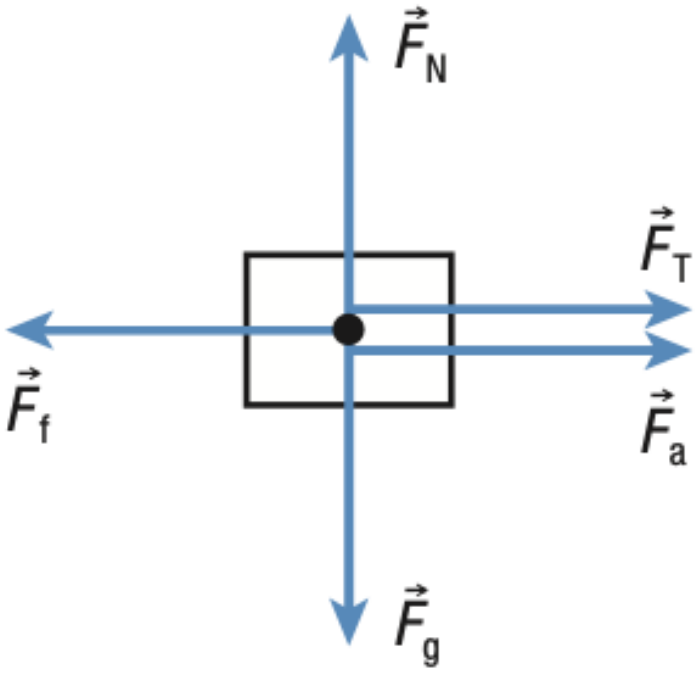
- Example: The child at the front pulls the rope, creating tension in it.
- Symbol: $\vec{F}_T$
- Direction: Along the rope, away from the object.

In a free-body diagram (FBD), both $\vec{F}_{app}$ and $\vec{F}_T$ start from the object and point forward.

Normal Force ($\vec{F}_N$)

The **normal force** is the perpendicular contact force a surface exerts on an object.

- Symbol: $\vec{F}_N$
- Always points **away from the surface.**
- Prevents objects from falling through the surface.



Friction ($\vec{F}_f$)

Friction resists the motion or attempted motion of an object.

- Symbol: $\vec{F}_f$
- Acts **parallel** to the surface.
- Direction: Opposite to motion or attempted motion.
- Example: If the wagon moves right, friction acts left.

Gravitational Force ($\vec{F}_g$)

Gravity (or the gravitational force) is the force of attraction between any two objects due to their mass.

- Symbol: $\vec{F}_g$
- Direction: Toward the center of Earth (downward).
- This is an **action-at-a-distance** force — it does not require contact.

How to Draw a Free-Body Diagram (FBD)

A **free-body diagram (FBD)** shows an object and all the forces acting on it. It helps visualize how different forces interact and predict motion.

Steps to Draw an FBD:

1. **Identify** all forces acting on the object. *(e.g., gravity, normal, friction, tension, applied forces)*
2. **Determine** the direction of each force. *(which way does each force act? up, down, left, right?)*
3. **Draw** a rectangle (the object) and arrows for each force. Label each arrow with the correct symbol (e.g., $\vec{F}_g$, $\vec{F}_N$). Arrows should point away from the object, and their lengths should show relative magnitudes.

Net Force ($\vec{F}_{net}$)

Once all forces are identified, they combine into a single **net force**.

$$\vec{F}_{net} = \sum \vec{F}$$

- The **net force** determines an object's acceleration:

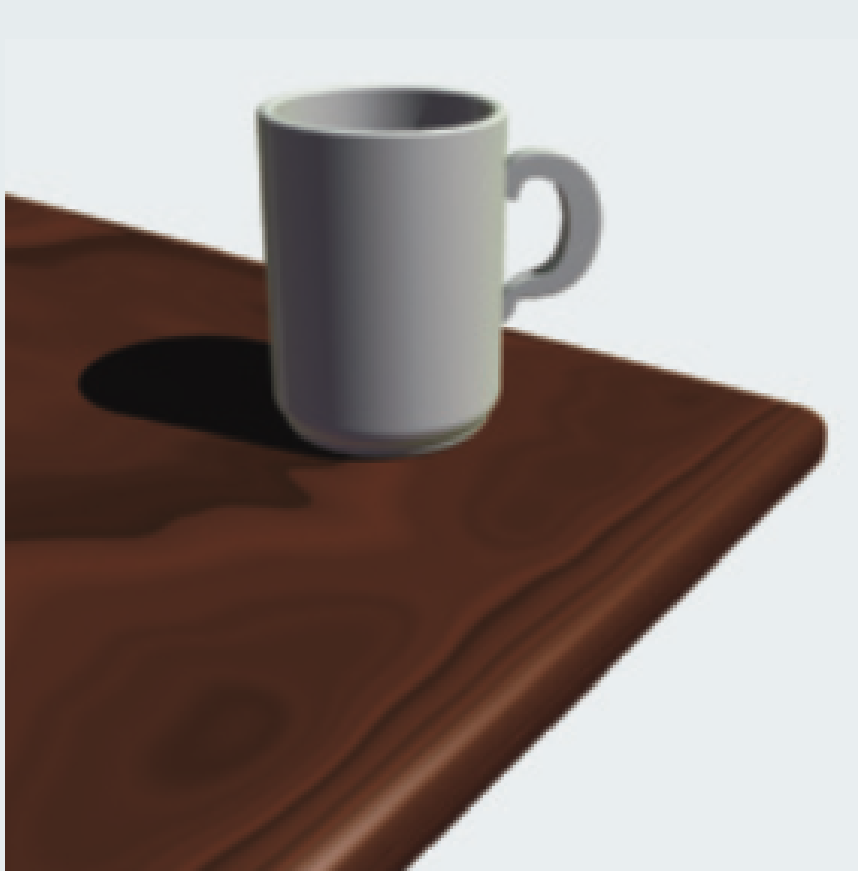
$$\vec{F}_{net} = m\vec{a}$$

- If $\vec{F}_{net} = 0$, the object stays at rest or moves at constant velocity.

Example 1: Coffee Cup at Rest on a Desk

Description: A coffee cup rests motionless on a flat desk. **Draw the corresponding FBD.**

System Diagram:



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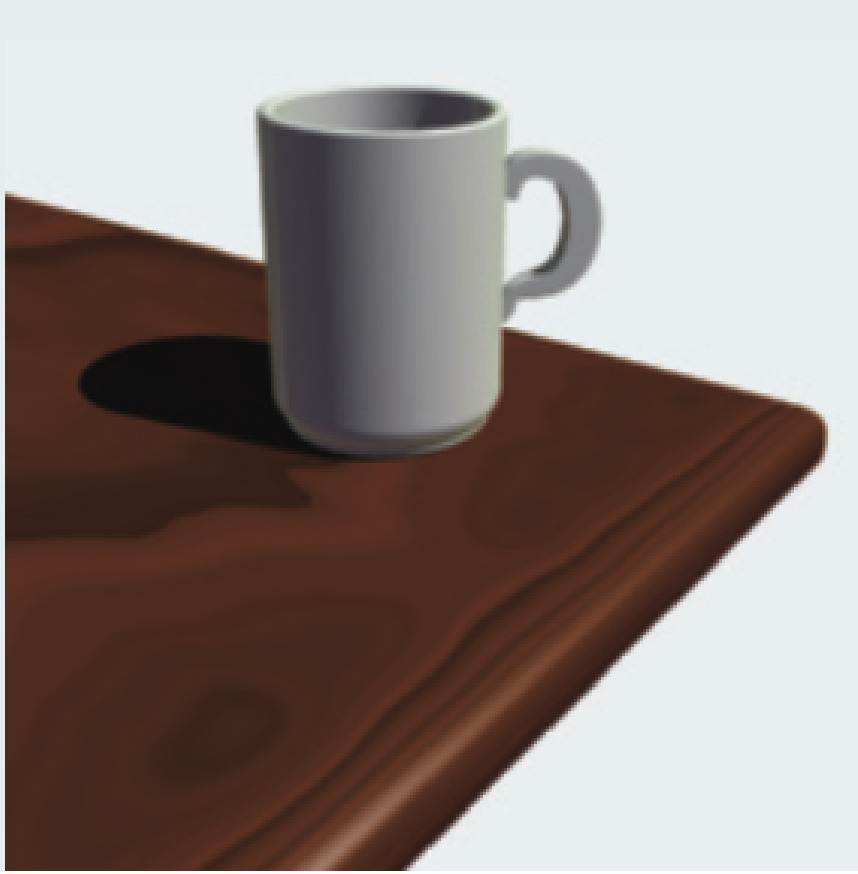

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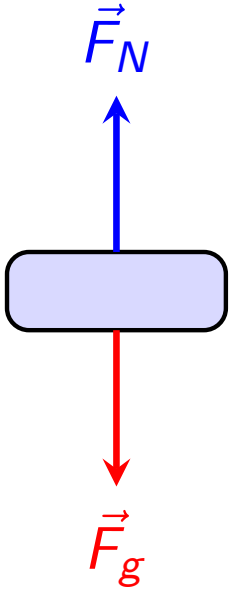

Example 1: Baseball Player Sliding

Description: George Springer has just put the Blue Jays in the lead with a home run in the 7th inning of Game 7 of the ALCS. Just as he's about to score, for the sake of grade 11 physics students, he decides to slide into home plate so they can analyze the forces acting on him.

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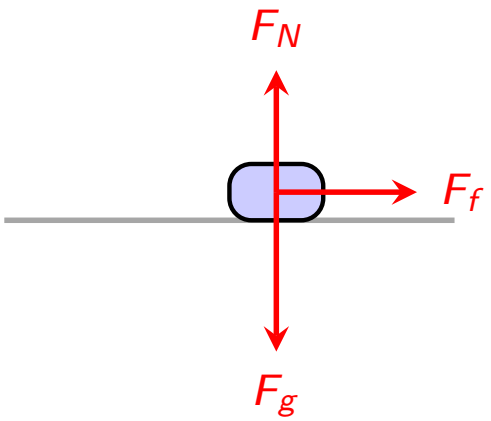

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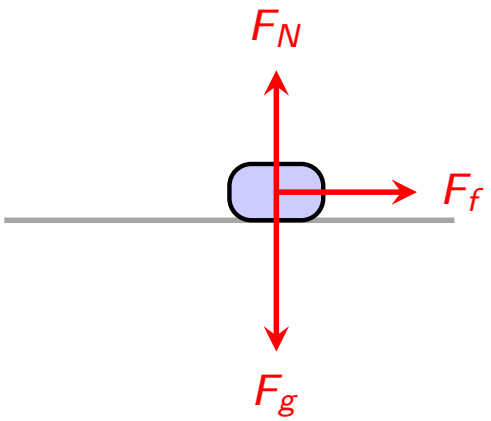

Note: There's no F_A because he's not continuously pushing on anything to keep moving.

Example 3: Determine the Net Force

Description: The floor exerts a normal force of 36 N [up] on a stationary chair. The force of gravity on the chair is 36 N [down]. Draw the FBD of the chair and use the FBD to determine the net force on the chair.

Step 1: Draw the FBD of the object.
Step 2: Identify which directions are positive.
Step 3: Add the forces on each axis to determine the net force.

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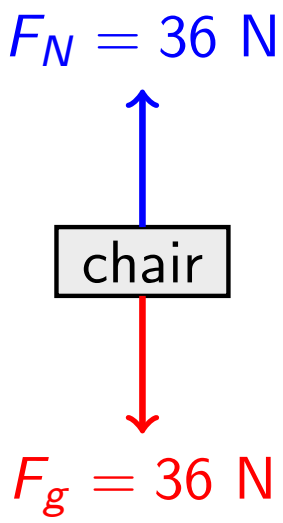

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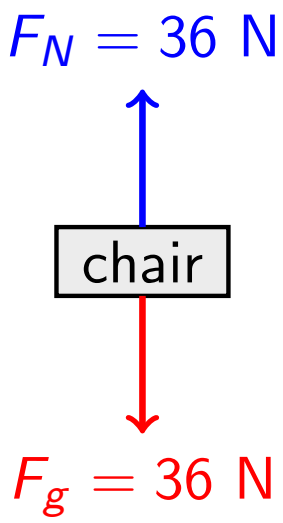

$$\vec{F}_{\text{net}} = \vec{F}_N + \vec{F}_g$$
$$\vec{F}_{\text{net}} = (+36\text{ N}) + (-36\text{ N})$$
$$\vec{F}_{\text{net}} = 0\text{ N}$$

Textbook Practice

Textbook Reference:
Pg. 114–121

Practice Questions:

- Pg. 120 #1–3
- *Forces and Free-Body Diagrams Practice* worksheet